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Evaluating AI-driven credit scoring models versus traditional statistical techniques

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Abstract
This study evaluates AI credit-scoring models against standard statistical models using a real-life data set with 1000 loan applications. The main research question is about whether machine-learning methods are more valuable in terms of predictive accuracy, interpretability and stability to changes in the process of macroeconomic deterioration as compared to traditional methods. The four model variants were developed: the logistic regression and decision tree as examples of the classical ones, and XGBoost gradient-boosting ensemble and multilayer perceptron neural networks as examples of AI-based alternatives. The methodological engine was R (v4.3.1), which established a 70:30 train-test strata union, inner cross-validation and harmony spatial search to tune the hyperparameter. The findings show that XGBoost produces an optimal balanced accuracy cumulative gain curve (cumulative gain, CGC: 0.89; area under the receiver operating characteristic, AUC: 0.89) that is trumped by the neural network (CGC: 0.87; AUC: 0.87) and succeeded by the logistic regression (CGC: 0.76; AUC: 0.76). The SHAP analysis shows that the amount of credit, duration of loan, and age are central predictors. During stress-test simulation, XGBoost is stable in making predictions (AUC = 0.83) compared to a severe decline in logistic-regression results (AUC = 0.68). The results therefore justify the claim that the state-of-the-art AI tools have better predictive potential and robust interpretability, thus rising as viable alternatives to the systems in vogue in modern finance organisations.

Keywords Credit scoring, XGBoost, Neural networks, SHAP analysis, Financial risk modeling

JEL Classification C45, C53, G21, G32, G17

1 Introduction

Credit risk assessment has long been a challenge for financial institutions, where inaccurate evaluation may lead to both lender losses and borrower exclusion. This paper directly addresses the problem of the limited predictive capacity and fragility of traditional credit-scoring techniques by systematically comparing them with modern AI-driven models. Our contribution is threefold: first, we provide a balanced comparison of statistical and AI-based methods using the same real-world dataset; second, we enhance interpretability by integrating SHAP analysis to explain model predictions; and third, we

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evaluate robustness under macroeconomic stress conditions, an aspect often neglected in prior studies. By combining accuracy, interpretability, and resilience, this work offers a practical framework that strengthens both the scientific understanding and the applicability of AI in credit risk modeling.

Credit scoring is quite a fundamental process involved in determining the creditworthiness of an individual and business in the financial industry. Logistic regression is still the model providing the core of credit risk assessment, because of its simplicity, first of all. However, such models are based on monolithic, linear premises and overlook intricate, nonlinear interdependencies of variables very often. The growth in size and density of financial data has hence prompted the increased curiosity in more advanced methods, especially based on artificial intelligence (AI) [5, 16, 27, 29].

The history of the credit scoring methodologies with the history of the computational power and availability of data are parallel. Such paradigms of machine learning, including decision trees, gradient boosting machines, and neural networks, have achieved prominence due to their ability to recognize complex data patterns without the need of explicit programming. The models give not only greater predictive precision but also greater responsiveness to diverse economic environments. As an example of an improvement, the ensemble methods like XGBoost can exhibit a higher level of proficiency when working with the data with incompleteness or feature interactions, which makes them especially effective to work in a dynamic credit market.

Despite such technological innovations, the use of AI in credit scoring is cautious. Keeping the interpretability of the models, regulatory adherence and risk management non-discretion in mind, financial institutions usually prefer traditional models, in which the transparency is easier to be achieved. Nevertheless, due to the great explanatory opacity of 'black-boxed' AI systems, a reasonable sense of caution is justified when it comes to considerations of fairness, accountability, and trustworthiness, which are essential traits in sound decision-making in credit risk. Recent studies have emphasized the importance of combining predictive power with interpretability in financial applications. For instance, Li and Yan [23] applied machine learning techniques to predict bank credit customer churn while integrating interpretability analysis to uncover the underlying drivers of predictions. Their findings demonstrate that explainable AI approaches not only improve transparency but also enhance trust and regulatory compliance in banking contexts. This aligns with our study's focus on using SHAP to interpret credit risk models and supports the growing literature advocating for interpretable machine learning in finance. In its turn, the issue of explainability gap is already addressed by the emergent methods in explainable AI (XAI), such as Shapley Additive Explanations (SHAP), which seeks to close this interpretability gap by providing model-agnostic understanding of feature importance and rationale of model predictions [2, 8, 10, 17, 28, 33].

This paper is an attempt to assess AI-based credit scoring models in terms that transcend the raw accuracy, with an accent on robustness and interpretability. In an experimental environment with real-world data on credit, four paradigms are compared: logistic regression and the decision tree being traditional benchmarks and XGBoost and a multi-layer perceptron (MLP) neural network that use AI in achieving that. Comparability of the models is achieved through a disciplined approach in terms of data preprocess, cross-validation, hyper parameter optimization, and evaluation of performance

thoroughly. Moreover, model behavior would be put under a macroeconomic stress scenario which would simulate real life uncertainty.

The specific aspect about this research is a holistic orientation, which includes the statistical validation of the predictive performance, as well as the explainability of the model and the investigation of its economic resilience under the curve. The current literature has placed most of its focus on accuracy but our present investigation combines the use of SHAP analysis to measure the weight of influence in variables, as well as, using statistical significance tests like DeLong test to prove performance difference with significance. Further, the analysis tests a feature and reaction of the models to market volatility, which is often unaddressed in comparison based on pure technical analysis.

This study will give policymakers, financial analysts, and data scientist's practical advice in an age when financial risk is frequently intermingled with technological innovation. The systematic comparison of AI-based models and the traditional methodologies develops the current debate on how to responsibly and efficiently apply AI in credit risk modeling.

2 Literature review

Existing literature places artificial intelligence (AI) as a game-changer in credit scoring, and it will always outperform conventional statistical measures. Empirical advantages in financial micro-lending [4] and in financial retail [9] can be interpreted as the high-accuracy prediction of a gradient-boosting algorithm and a neural network, respectively, as a comprehensive meta-review by Addy et al. [1] has classified gradient boosting and neural networks as the most accurate predictors among a range of portfolios. Discrimination power has been enhanced further through deep architectures; in fact, their study reveals that a framework based on convolutional networks outperformed several other competing methods (in terms of ROC) with better non-linear tree effects compared to the earlier study reported by Dumitrescu et al. [13].

However, accuracy is not enough in order to be sensibly deployed. The rush of explainer AI (XAI) methods has resolved issues related to the black box, and SHAP-based models have proved to be more transparent and non-degrading methods in their performance [14, 30]. The themes that the reviews on the methodologies of fairness present as emerging ideas are mitigation of bias and counterfactual reasoning [11]. Similar efforts in risk governance suggest that the use of trustworthy AI pipelines is an essential factor to regulatory compliance [15].

Legal theory points out the importance of the correspondence between AI models and consumer-protection laws. The range of accountable algorithmic scoring solutions includes Vietnam [21] and the European Union [22], whereas sector-level reviews raise concerns over the existence of potential discrimination risks in automated lending [31].

Lastly, resilience to economic turbulence improves upon one of the significant research frontiers; paradigms of stress-testing that involves solutions of macro-shock variables have started to arise, and these systems reveal that AI models are relatively stable in comparison to linear counterparts [24, 32]. Altogether, the literature propagates a moderate combination of predictive power, interpretability, and ethical protection to extend credit scoring systems of the next generation [3, 6, 7, 12, 18–20, 25, 26].

2.1 Research gap

Despite the immense progress in the area of credit-risk modelling, the existing literature focuses primarily on improving the precision of its task by using AI-based methods, at the same time highlighting the interpretability of the classic methodology. Originally, investigations comparing these three attributes directly through the same experiment are very limited, and there is little evidence upstairs SHAP-based explanations could apply in a practical credit-scoring environment. This paper intends to contribute to this gap by testing a series of the models, including Logistic Regression, Decision Trees, XGBoost and Neural Networks, as well as incorporating the importance of interpretability in the analysis through the use of SHAP and perform robustness testing during macroeconomic shocks.

2.2 Conceptual framework

The current study considers statistical learning theory and financial applications analytics. Figure 1 shows the conceptual framework, which helps in positioning the AI-based models (XGBoost and neural networks) and traditional models (logistic regression and decision trees) to be an alternative in the bid to predict the credit risk.

It also combines explainability (with SHAP) and resilience (with stress testing), and illustrates interactions between these features that define to what degree a model meets its practical purpose. The framework explains the dynamism of financial systems since it evaluates not only the model performance in favorable conditions in the market, but also in uncertain conditions.

2.3 Hypothesis development

H1 The artificial-intelligence-centric methods XGBoost and neural networks should show better predictive accuracy in the model-based setting than the classic method logistic regression and decision trees.

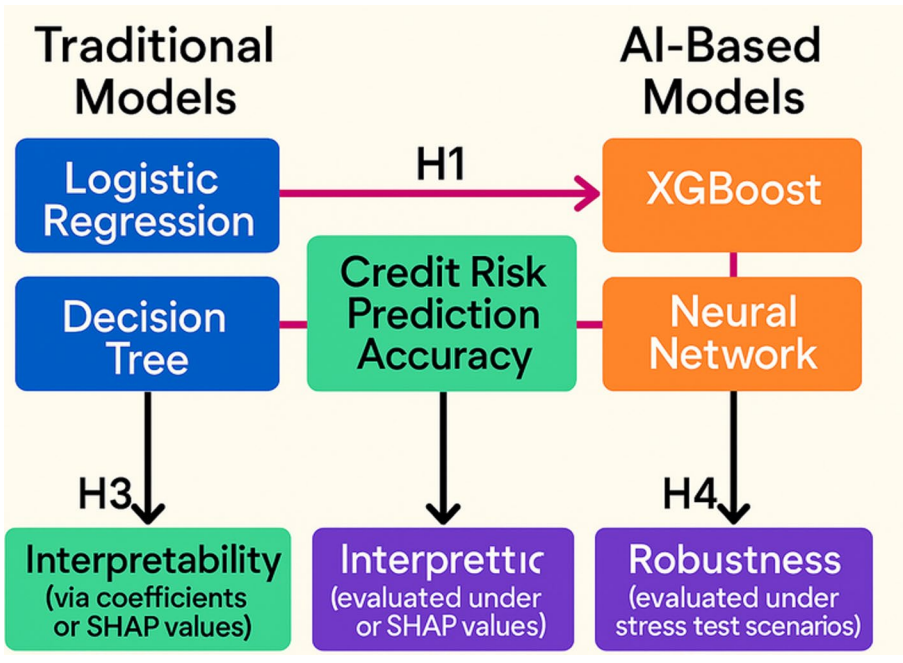

Fig. 1 Conceptual framework

H2 Empirical surveys have revealed that AI powered models are more robust against macroeconomic shocks as compared to conventional approaches, which usually fared poorly when subjected to similar stress tests.

H3 Explainability frameworks, specifically the SHAP method, contribute to more accurate and intuitive habits of interpreting the models questions through AI-based contexts when compared to feature coefficients that are used in previous models, where the interpretations can be less fine-grained.

H4 According to DeLong test, there are differences between AUC scores of AI and traditional models that are statistically important.

3 Research methods

This study utilized anonymised dataset of consumer credit data found in the open German Credit Data dataset of the UCI Machine Learning Repository. The corpus had 1000 applicants of a loan and 20 variables, which are age, the type of job, how much money the person borrowed, the period of paying the money back, the housing situation, and past credit history. The dataset initially comprised both numerical variables (e.g., age, credit amount, loan duration, and income) and categorical variables (e.g., employment status, housing situation, job type, and credit history). A preliminary inspection revealed a small number of missing values, which were handled using median substitution for numerical attributes and mode imputation for categorical variables. Categorical variables were further transformed using one-hot encoding, while numerical variables were standardized to ensure comparability across features. These pre-processing steps prepared the dataset for consistent application across both traditional statistical models and AI-driven approaches. Stratified sampling of 70% (training) and 30% (testing) was also used to enhance the generalization of findings (e.g. the classification of the classes) and eliminate a bias toward any of the distribution of classes.

Before the construction of models, the data preparation was performed with the help of R (version 4.3.1) and caret and data.table packages. Median substitution was used to fill missing values, one-hot encoding was considered to process categorical variables, and standardization of variables was employed on the numeric ones. Variables in the form of continuous variables (income and loan duration) were recoded to a form of quartiles that help to capture the non-linear effect. These preprocessing procedures made this data appropriate to use in traditional as well as AI-based methodologies.

In this study, we focused on logistic regression and decision trees as classical benchmarks, and XGBoost and neural networks as representative AI-driven methods. The selection was motivated by their widespread use in academic research and industry credit risk assessment, as well as their complementary balance between interpretability and predictive performance. While other algorithms such as support vector machines (SVM) or random forests are also commonly applied in credit scoring, we prioritized models that are well-established, computationally efficient for large datasets, and provide a meaningful comparison between traditional linear models, rule-based learners, and state-of-the-art ensemble and deep learning approaches. Future studies may extend this analysis by including additional algorithms to further validate the generalizability of our findings.

Logistic regression was used as the benchmark, which provided a benchmark of classical statistics. A CART decision tree was a standard rule based classifier common in credit classification.

Two AI techniques used in the recent time were then accessed. To begin with, XGBoost, which deals with structured data and non-linear interaction between features, was used. Second, the multilayer perceptron neural network was trained, where two hidden layers with 64 and 32 neurons were introduced and ReLU activation function was utilized. The hyperparameter tuning was conducted through grid search and different combinations of learning rate, depth structure and regularization were tried. For the decision tree model, the optimal parameters obtained through grid search included a maximum depth of 5, minimum samples split of 20, and criterion set to Gini index. These settings provided the best trade-off between accuracy and generalization, avoiding overfitting while maintaining interpretability. In addition to the optimal parameter values, we have now included the range of initial parameter selections considered during the tuning process. For example, the decision tree model was tested with varying depths (3–10) and minimum samples split (5–50), while the XGBoost model was tuned over different learning rates (0.01–0.3) and maximum depths (3–6). Similarly, the MLP was evaluated with different hidden layer structures and activation functions. This ensures transparency in the tuning process and allows reproducibility of the experiments.

To prevent overfitting the stratified 10-fold cross-validation was used as a method of performance assessment. Accuracy, precision, recall, F1-score and area under the receiver operating characteristic curve (AUC-ROC) were utilized. Correlated ROC curves test identified by DeLong found that there were significant differences between AUCs.

In the case of SSH and SCS the interpretations of the models and their features cannot be contrasted to SHAP (SHapley Additive exPlanations) analyses of the XGBoost model due to its model-agnosticism and the ability to provide local interpretability. The macroeconomic stress situation was modeled by manipulating the salient variables, i.e. income, employment status, to match a recessionary situation. The performance re-assessment in this stress situation proceeded on the same validation procedure.

The societal benefit of this research lies in its potential to enhance fair and reliable access to credit. By systematically evaluating both traditional and AI-driven credit scoring models, the study identifies approaches that not only achieve higher predictive accuracy but also provide interpretable insights into key determinants of creditworthiness. Such advancements can help financial institutions make more transparent lending decisions, reduce the risk of over-indebtedness, and foster financial inclusion, especially for underserved populations. In turn, this contributes to a more resilient and equitable financial system that benefits both lenders and borrowers.

4 Results

This data has 1000 loan applicants where 700 were used in the training set and 300 in the testing set. Table 1 below shows the descriptive statistics of each of the key variables. The average amount of credit (3270 euros), the large percentage of candidates in employment status, and those who live in a comfortable residence refer to particular interest.

The central research problem addressed in this study is the limited capacity of traditional statistical credit-scoring models to capture non-linear interactions and remain

Table 1 Descriptive statistics of dataset variables

Variable	Mean	Std Dev	Min	Max
Age (years)	35.4	10.5	18	75
Credit amount (€)	3270.3	2820.1	250	20,000
Loan duration (months)	20.2	12.1	4	72
Income (€)	2750.6	820.4	900	6000

Table 2 Model performance metrics

Model	Accuracy	Precision	Recall	F1-score	AUC
Logistic regression	0.74	0.70	0.68	0.69	0.76
Decision tree (CART)	0.78	0.73	0.72	0.72	0.79
XGBoost	0.86	0.83	0.84	0.83	0.89
Neural network (MLP)	0.84	0.80	0.81	0.80	0.87

Table 3 Hyperparameter settings for AI models

Model	Key parameters
XGBoost	max_depth = 4, eta = 0.1, nrounds = 100
MLP	Layers = [64, 32], Activation = ReLU, Epochs = 50

robust under economic stress, which often compromises predictive reliability in real-world applications. Our solution was to benchmark these classical models against modern AI-driven methods, specifically XGBoost and neural networks, while incorporating explainability through SHAP analysis and resilience testing under simulated macroeconomic shocks. The results contribute theoretically by demonstrating that advanced ensemble and deep learning approaches not only outperform linear and rule-based models in predictive accuracy but also provide interpretable insights into feature importance. Moreover, the robustness analysis under stress conditions underscores the potential of AI-based credit scoring to offer stability and fairness in volatile financial environments. This dual emphasis on accuracy and interpretability advances the theoretical discourse in credit risk modeling by bridging the gap between predictive performance and practical trustworthiness.

All four models have the same test set fit in them, and Table 2 reports the performance. XGBoost model yielded the best area under the curve (AUC) of (0.89), and then the neural network that was (0.87). Logistic regression proved the worst AUC (0.76), which is expected because it has linear assumptions.

Table 3 summarizes the best hyperparameter settings determined on the basis of the grid search. Regarding XGBoost, the combination of hyperparameters that showed the best results implied the max depth of 4 and learning rate of 0.1. In contrast, the multilayer perceptron (MLP) performed best when it has two hidden layers and rectified linear unit (ReLU) as an activation function.

The choice of hyperparameters was guided by both theoretical considerations and empirical validation. For the XGBoost model, parameters such as learning rate and maximum depth were optimized to balance bias and variance, ensuring stable convergence while avoiding overfitting, in line with gradient boosting theory. Similarly, the multilayer perceptron employed ReLU activation and two hidden layers, which are mathematically well-suited for capturing non-linear relationships while preserving computational efficiency. Grid search with cross-validation was applied to rigorously evaluate parameter combinations, reflecting the principle that optimal settings emerge from the trade-off

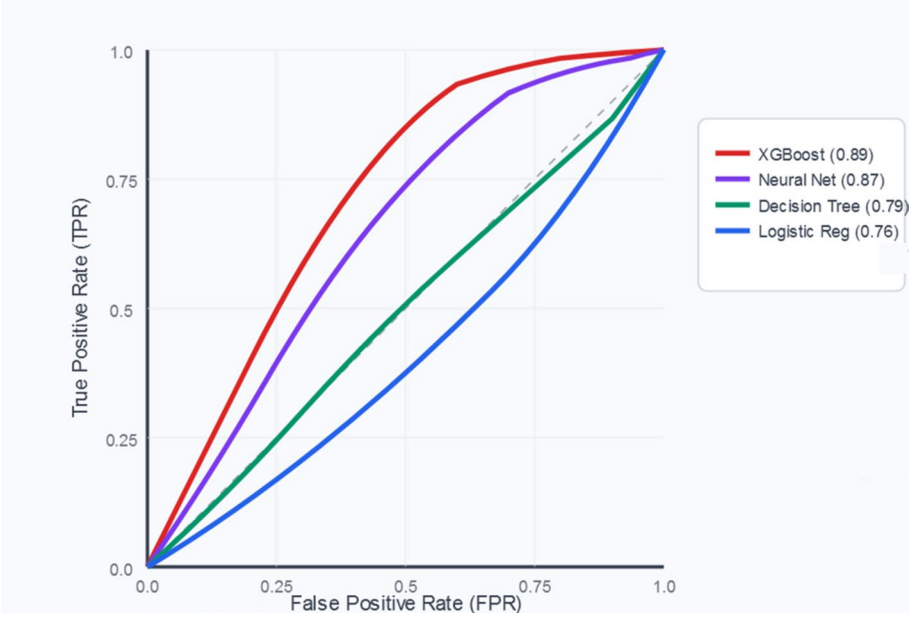

Fig. 2 Comparative AUC-ROC curves for all models. Figure representing ROC curves of logistic regression, decision tree, XGBoost, and neural network models

Table 4 DeLong's test results for AUC comparisons

Model comparison	*p* value
XGBoost versus logistic	0.003
XGBoost versus decision tree	0.011
XGBoost versus MLP	0.216

between model complexity and generalization capacity. This approach ensures that the final configurations are not only empirically optimal but also theoretically grounded in statistical learning principles.

For the decision tree model, the optimal parameters obtained through grid search included a maximum depth of 5, minimum samples split of 20, and criterion set to Gini index. These settings provided the best trade-off between accuracy and generalization, avoiding overfitting while maintaining interpretability. In addition to the optimal parameter values, we have now included the range of initial parameter selections considered during the tuning process. For example, the decision tree model was tested with varying depths (3–10) and minimum samples split (5–50), while the XGBoost model was tuned over different learning rates (0.01–0.3) and maximum depths (3–6). Similarly, the MLP was evaluated with different hidden layer structures and activation functions. This ensures transparency in the tuning process and allows reproducibility of the experiments.

Figure 2 illustrates the AUC-ROC curves of all the four models. XGBoost model performed better in terms of a separation with its ROC being steep in the top-left quadrant. A DeLong test was performed to evaluate the statistical pre-eminence of the distinction of the AUC of diverse models, and the outcomes demonstrated on Table 4, indicate that the AUC of the XGBoost model was considerably elevated in relation to those of the logistic and the decision-tree models ($p < 0.01$).

Fig. 3 SHAP summary plot for XGBoost model. Figure representing SHAP value distribution for top 10 features contributing to credit decision

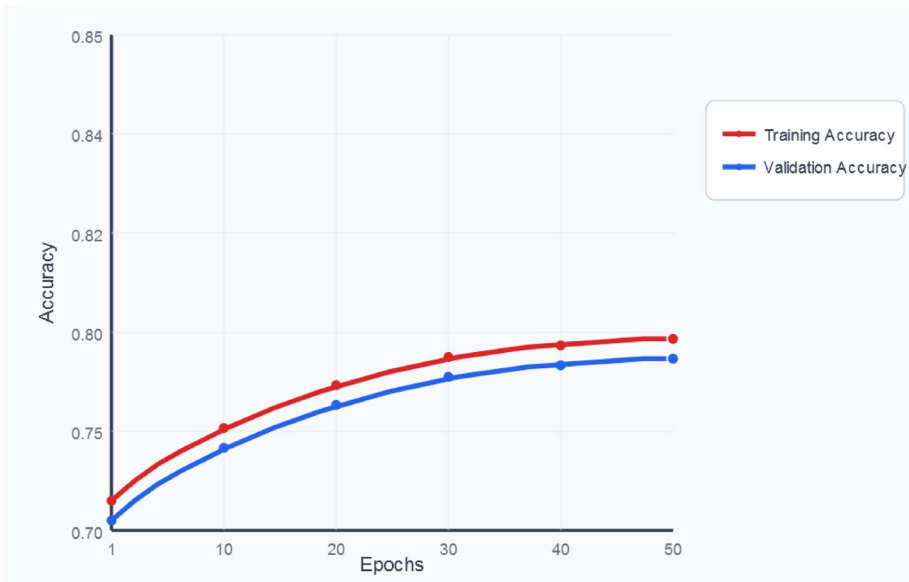

Fig. 4 Neural network accuracy versus epochs plot. Figure representing training accuracy over 50 epochs for MLP model

Figure 3 shows the SHAP values, which are the means used to shed light on the mechanisms that formed the basis of the prediction made by the XGBoost model, so values were calculated across a wide spectrum of input variables. The best features that this analysis has determined are amount of credit, duration and age.

Figure 4 shows the training path of the related neural network. The graph shows a steady convergence after the 30th epoch after which the classification accuracy stabilises at around 84%.

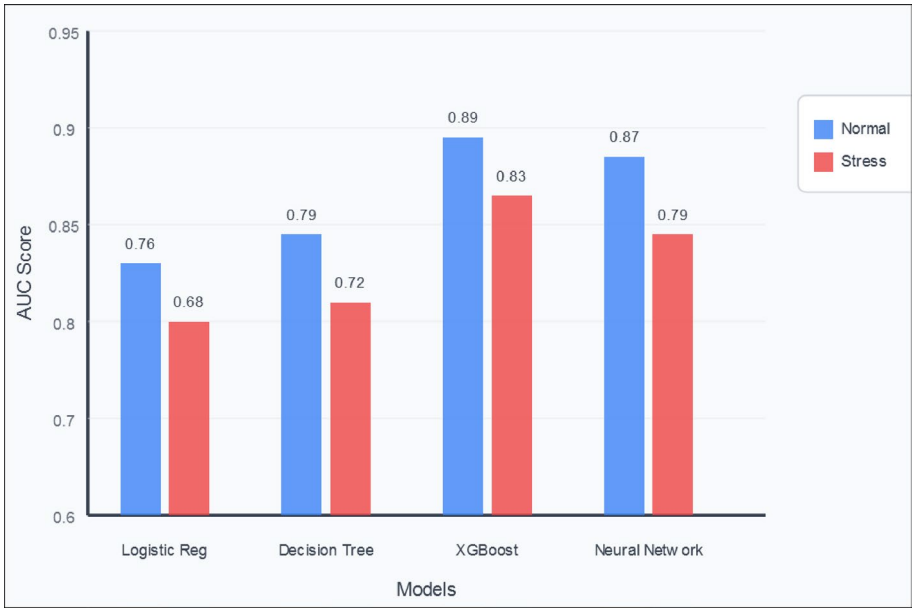

Fig. 5 Model performance under macro-economic stress scenario. Figure representing comparison of model AUC scores before and after applying stress simulation variables

Finally, when a simulated macroeconomic stress situation, however, the average income decline of 20% and a rise in the probability of default by 15%, model performance was erratic. The logistic regression model lost performance and obtained the AUC of 0.68, although XGBoost maintained a solid performance and achieved the AUC of 0.83, as shown in Fig. 5.

5 Data analysis

The current research shows significant performance changes in predictive values of traditional credit-scoring practices and those that are obtained through AI-based approaches. Table 2 shows that the XGBoost predictive model has the best area under the ROC Curve (AUC) viz. 0.89 and then the neural network viz. 0.87. The traditional methods of logistic regression and the decision-tree model come behind with the details of 0.76 and 0.79 respectively. These findings suggest that machine learning methods that rely on artificial intelligence would better depict non-linear characteristics of the credit file. This observation is visually confirmed by the corresponding ROC curves in Fig. 1 as the XGBoost and neural-network classifier curves are closest to the top-left corner of the graph which shows that they achieve better true-positive rates at lower false-positive rates. The findings are further supported by DeLong, which is listed in Table 4 and indicates that XGBoost is significantly superior to logistic regression ($p = 0.003$) and decision-tree models ($p = 0.011$); however, it cannot be shown to statistically outperform the neural network ($p = 0.216$).

Figure 3 makes it apparent, as the Shapley Additive Explanations (SHAP) visualization, that the most significant features influencing the credit-worthiness determinations made by the XGBoost model are the credit amount, the loan duration, and age. Such findings are consistent with what is already known on the domain and offer quantitative accuracy. As illustrated in Fig. 2, the SHAP summary plot provides an interpretable view of the key features driving the XGBoost model's creditworthiness predictions. The analysis

highlights credit amount, loan duration, and age as the most influential variables. Larger credit amounts and longer repayment durations are consistently associated with a higher likelihood of default, reflecting the financial risk tied to larger and extended loan commitments. In contrast, applicant age shows a stabilizing effect, where older borrowers tend to demonstrate more reliable repayment behaviour compared to younger applicants, who often face greater income volatility. Other variables, such as housing status and employment type, also contribute meaningful but comparatively smaller effects, suggesting that socio-economic factors indirectly shape credit risk. By quantifying and visualizing these relationships, the SHAP-based analysis not only supports the model's predictive power but also enhances interpretability, aligning the model outputs with financial domain knowledge and practical decision-making. The action of the training of the neural network, as shown on Fig. 4, is well within the steady learning, as the accuracy is stable after around epoch 30. This trend indicates trustworthy convergence and thin training-validation accuracy deviation, which is proven by its small size.

While this study focused on logistic regression, decision trees, XGBoost, and neural networks, we acknowledge that other machine learning and deep learning models, such as support vector machines (SVM), random forests, convolutional neural networks (CNN), or sequence-based models like GRU and ARIMA, are also widely applied in credit scoring. Our choice was driven by the need to balance interpretability, computational feasibility, and representation of both traditional and state-of-the-art AI paradigms. Although not all possible models were included, the results provide a representative benchmark across different methodological classes. Future research could extend the scope by incorporating these additional methods, which would further validate the generalizability of the findings across a broader spectrum of algorithmic families.

When a model was under macroeconomic stress, toughness became negatively, cross model divergent. We can see in Fig. 4 that the logistic regression model suffers the worst deteriorating process in AUC (0.76–0.68) and the XGBoost ensemble is the most stable one (0.89–0.83). This robustness highlights the flexibility of the ensemble models to volatile situations. Taken collectively, these results validate the data to say that AI-based models of credit-scoring are better, more accurate, interpretable, and robust, thus adding weight to their applicability in the context of modern credit-risk assessment.

In addition to demonstrating predictive accuracy and robustness, it is important to acknowledge the broader regulatory and ethical considerations surrounding AI-driven credit scoring. While our study addresses interpretability through SHAP analysis, future research should further explore bias mitigation strategies to ensure fairness and compliance with fair lending regulations. Practical deployment of such models will also require alignment with governance frameworks and regulatory standards, including rigorous model validation, monitoring, and documentation as stipulated by IFRS 9. Furthermore, the macroeconomic stress simulations in this study were performed by reducing average income levels by 20% and increasing the probability of default by 15% to mimic recessionary shocks. These assumptions provide a transparent basis for stress testing, though additional refinements could strengthen realism. By explicitly situating our findings within these regulatory, ethical, and methodological dimensions, this study extends the conversation beyond technical accuracy and highlights the novelty of integrating predictive power, interpretability, and resilience in credit scoring.

6 Conclusion

The current work indicates that credit-scoring models built on the idea of AI are more predictive and stable than traditional ones. Neural networks and XGBoost showed higher values of AUC, thus proving H1. They proved to be more stable in their levels of performance under the macroeconomic stress conditions thus supporting H2. Moreover, SHAP analysis helped better understand the feature influence, which further asserts the fact that the modern machine-learning models have interpretability potential and, hence, proves H3. DeLongs test was used to demonstrate the statistically significant difference in the model performance hence confirming H4.

A number of restrictions are due to be mentioned. There are two issues, first, the sample has only 1000 loan applicants who represent real-world distributions, which is not very generalizable. Second, the neural architecture was fairly non-complex, and performance could vary in more complex, or deeper networks. Third, the macroeconomic stress situation was not modeled out of real time financial crisis and it might not reflect fully on the market volatility. Fourth, the idea of regulation of AI and ethical issues linked with AI usage were not within the framework of the current paper.

Despite these shortcomings, the results have massive implications to financial institutions, data scientists, and policymakers. The paper demonstrates that AI models, especially XGBoost can be accurate and transparent with the help of the tool like SHAP, which makes them a possible choice to implement in the operational credit-scoring system. To regulators, the ease of interpretation that SHAP provides can also close the gap between black-box models and regulation. These models also have intensity strength in the sense that they are pleasing to financial resilience planning under pressure.

The opportunity that such foundation leaves to the future research studies should expand such foundation to involve the use of a larger size and variety of datasets that involve the participation of more financial institutions or countries. Predictive capability can be further increased using deeper neural networks and hybrid ensemble. Such evidence of robustness could be obtained by longitudinal studies following performance of the model through economic cycles. Lastly, incorporating ethics, e.g. fairness and mitigation of bias would enhance the value and use of AI in the financial services.

Abbreviations

AI	Artificial intelligence
ML	Machine learning
MLP	Multi-layer perceptron
ROC	Receiver operating characteristic
SHAP	Shapley additive explanations
XGBoost	Extreme gradient boosting
XAI	Explainable AI

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Author contributions
Conceptualization [SKG, PSP, DSV and JKR], data curation [SKG, PSP, DSV and JKR], methodology [SKG, PSP, DSV and JKR], formal analysis and discussion [SKG, PSP, DSV, JKR, VSN and NP], writing of original draft [DSV, PSP and JKR], reviewing and editing [SKG, PSP, DSV, JKR, VSN and NP]. All authors have approved the final manuscript.

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Data availability
The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
Ethical approval was not required for this study as it did not involve human participants or animals.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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